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Temporal Remote Sensing and GIS Analysis of Land Use/Land Cover Transitions: A Case Study in Annamayya District, Andhra Pradesh, India

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Abstract: Understanding and managing the changing landscape of our planet requires analyzing land use and land cover (LULC) transitions. This study combines remote sensing data and GIS analysis to uncover the intricate patterns and drivers behind these transitions. By integrating GIS analysis, we identify factors such as urbanization, agricultural expansion, deforestation, and natural resource management contributing to these changes. Accurate data preprocessing and calibration are emphasized to reduce errors and uncertainties in LULC maps. The study covers the period from 2017 to 2022, utilizing digitized LULC maps created with GIS tools and satellite data interpretation, specifically Sentinel-2 images. The landscape was classified into six land cover types: agricultural land, built-up areas, barren or degraded land, flooded vegetation areas, forests, vegetation or shrub-covered areas, and waterbodies. Findings revealed a decrease of 1063.3 km² in barren land between 2017 and 2022. Agricultural land expanded by 228.93 km², forest land increased by 632.90 km², and waterbodies grew by 33.05 km². These changes suggest a conversion of areas likely influenced by the high intensity of rainfall between 2017 and 2022, leading to notable ecological consequences such as reduced soil erosion and improved biodiversity protection. The study's results have significant implications for land management, environmental protection, and sustainable development. The extensive analysis of spatial and temporal data equips policymakers, urban planners, and researchers with crucial knowledge to make well-informed decisions. This research provides a solid basis for monitoring and managing LULC transitions, offering valuable methodologies and insights to promote a resilient and harmonious coexistence between human activities and the environment.

Key words: land use/land cover; remote sensing; GIS; transitions; environment; sustainable development

1 Introduction

The study of LULC changes has received considerable scholarly interest because of its substantial effects on ecosystems, climatic patterns, and socio-economic structures. The use of remote sensing and GIS has become essential in the field of monitoring and comprehending these transitions.

These techniques enable researchers to examine the temporal and geographical patterns of changes occurring in various landscapes (Areendran et al., 2013; Butt et al., 2015; Rizk and Rashed, 2015). This literature review examines the significant contributions made by prior research in the domains of temporal remote sensing, GIS analysis, and LULC

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transitions, therefore providing a foundation for the publication titled “Temporal Remote Sensing and GIS Analysis of LULC Transitions” (Pattanaik et al., 2011; Ayele et al., 2018; Langat et al., 2021). The study of remote sensing data across time has seen significant development, progressing from single-date studies to more sophisticated multi-temporal techniques that effectively capture and represent dynamic changes. The importance of time-series data in the identification and characterization of LULC changes is emphasized in the research conducted by Lu et al. (2014) and Dutta et al. (2018). The usefulness of long-term datasets in capturing progressive and abrupt transformations is shown by the usage of Landsat and Sentinel satellite images, as evidenced in the studies conducted by Roy et al. (2017) and Verburg et al. (2020). A variety of change detection techniques have been devised for the analysis of LULC transitions, including both pixel-based and object-based methodologies. The first research conducted by Lu et al. (2004) demonstrated the effectiveness of post-classification comparison. Subsequent studies by Defries et al. (2010) have further advanced the field by using more sophisticated methodologies, including spectral mixture analysis and decision trees. The use of machine learning methods, as shown by Clewley et al. (2014), has enhanced the precision and effectiveness in identifying changes in land cover. The utilization of GIS in comprehending LULC transitions surpasses mere change detection, as it encompasses the spatial study of the factors influencing these changes and their subsequent impacts (Pande et al., 2021; Hussain et al., 2022). The importance of spatial measurements in identifying urban development patterns has been highlighted in urban expansion research, as evidenced by the work of Angel et al. (2010). Furthermore, the examination of landscape fragmentation, as expounded upon by McGarigal and Marks (1995), has provided insights into the ecological consequences of LULC changes. The utilization of GIS methodologies has facilitated the examination of the spatial framework surrounding LULC transformations, as well as the identification of the fundamental factors driving these alterations. In their study, Wang et al. (2016) used spatial measures to evaluate the patterns of urban growth, emphasizing the significance of GIS in evaluating alterations in fragmentation and connectedness. Moreover, the integration of GIS with machine learning techniques, as emphasized by Li et al. (2019) and Yang et al. (2021), has resulted in improved precision in LULC categorization and detection of changes. The factors influencing LULC changes exhibit spatial and temporal variations. The authors Seto et al. (2013) highlight urbanization as a significant factor, whilst Lambin and Meyfroidt (2011) delve into the intricate relationship between changes in land use and the spread of agriculture. The necessity of monitoring and controlling LULC transitions, such as deforestation (Gibbes et al., 2018) and habitat loss (Hansen et al., 2016), is under-

scored by their significant environmental impacts. Despite the significant progress made in temporal remote sensing and GIS methods, there is still a lack of comprehensive research that combines these tools to fully understand the spatiotemporal dynamics of LULC shifts. The publication, produced by Rajasekhar et al 2020, addresses this gap in knowledge by providing a comprehensive examination of LULC changes. This study is conducted via the integration of multi-temporal satellite images with GIS approaches. The paper contributes to the current body of knowledge by addressing methodological problems and highlighting the significance of precise data processing. It presents a comprehensive framework for monitoring and comprehending the intricacies of LULC changes. Rajasekhar et al. (2018) and Kadam et al. (2021) are scholars who possess specialized knowledge in the fields of remote sensing, GIS analysis, and environmental science. The researchers have accumulated a combined study experience of more than ten years, covering a diverse array of studies pertaining to the dynamics of land use, landscape ecology, and spatial analysis (Rajasekhar et al., 2018a; 2018b; 2019c; 2021). The collective endeavours of the researchers have resulted in the creation of innovative approaches for examining LULC changes, specifically emphasizing the use of temporal remote sensing and GIS methods. In their work, Ramachandra et al. (2021) provides a thorough examination of the complex relationship of land use changes, temporal dynamics, and GIS spatial analysis, drawing upon their collective knowledge. The use of temporal remote sensing and GIS analysis in LULC research has significant implications for promoting sustainable development and effective environmental management. The incorporation of research findings, such as those shown by Verburg et al. (2015), may enhance the effectiveness of land-use planning and policy development. Verburg et al. (2015) underscored the significance of taking into account the socio-economic factors that underlie changes in land cover. These observations contribute to the optimization of resource allocation and the reduction of negative environmental consequences. Despite notable advancements, there are still persistent issues in the field. These challenges include the need for data with enhanced spatial and temporal resolution, the development of more effective classification algorithms, and the incorporation of social and economic elements into LULC models in a more comprehensive manner (Rajasekhar et al., 2019a, 2019b, 2020; Kadam et al., 2021). In addition, the emergence of cloud computing and big data analytics provide novel prospects for effectively managing the extensive volume of temporal remote sensing data. The integration of temporal remote sensing and GIS analysis has significantly transformed our comprehension of LULC changes. This amalgamation has provided us with a profound insight of the intricate dynamics between natural phenomena and human interventions (National and Pillars,

2010; Jiang and Tian, 2010; Li et al., 2011, 2021; Areendran et al., 2013; Munthali et al., 2019). The objective of this publication is to contribute to the expanding field of knowledge by offering comprehensive research that utilizes these approaches to investigate the temporal patterns and factors influencing LULC changes.

2 Study area

The Annamayya District is an urban region surrounded by substantial industrial and agricultural operations as well as lush forest. The research region is between latitudes 13°18'30"N and 14°32'04"N and longitudes 78°03'59"E and 79°28'16"E. The watershed has a semi-arid environment

with an annual rainfall of 680 mm and elevation ranging from 500 to 700 m. The northern half of the watershed, which is mostly underlain by mudstones, was formerly deforested and terraced for agricultural use. Forest patches formed as a result of natural afforestation after the end of agriculture. The major traditional economic activity in the region were cereal farming and sheep herding, while woods supplied locals with lumber and fuelwood. Terrace development resulted in an increase in cultivated area as a consequence of demographic pressure. Later, migration to more desirable industrial employment resulted in a significant population decline (Figure 1).

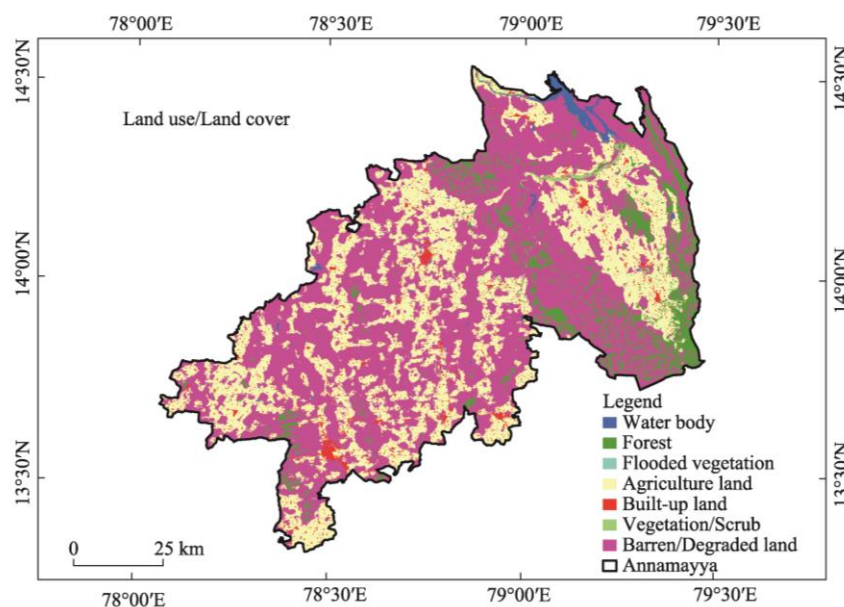


Figure 1 Location of Annamayya District, Andhra Pradesh, India

3 Materials and methods

The comprehensive procedure used in this examination of LULC dynamics using RS and GIS. The graphic representation showcases the primary stages of data gathering and preprocessing, image classification, change detection, and LULC classification. The research was carried out with ERDAS Imagine 2014 and ArcGIS 10.8 software applications (Rogan and Chen, 2004; Ā and Tateishi, 2007).

3.1 Data and pre-processing

The field of image processing involves the alteration and interpretation of digital images. The use of resolution merging is utilized to enhance the spatial resolution of images by the integration of images that possess differing pixel densities. In the context of remote sensing, radiometric enhancement techniques are used to improve the classification accuracy of area images by addressing common challenges such as stripping and banding errors that occur due to misalignment of the detector. Moreover, the use of principal

component analysis (PCA) in image visualization is advantageous due to its ability to compress data, provide uncorrelated output bands, isolate noisy components, and reduce the overall complexity of data sets (Kandrika and Roy, 2008; Ayele et al., 2018; Hao et al., 2021; Pande et al., 2021). During the preprocessing stage, Sentinel 2 images with a resolution of 10 meters underwent several changes. The process involved in this study included resampling techniques to modify spatial and spectral resolutions, layer stacking to combine many scenes, and the application of geometric changes. The manipulation of pixel arrangement in digital images, known as geometric transformation, plays a crucial role in facilitating subsequent processing tasks (Kandrika and Roy, 2008; Suneela and Mamatha, 2016; Langat et al., 2021). The process of resampling photographs entails the transformation of satellite images from a higher level of detail to a lower spatial resolution. This may include visual representations derived from diverse satellite sensors exhibiting distinct levels of resolution. The choice of the

resampling technique is contingent upon several aspects, including the relationship between the dimensions of the input and output pixels, as well as the intended use of the resampled image.

The closest neighbour approach was used to resample the Sentinel 2 images in this investigation. The decision was taken to preserve the radiometric information of the original image. Furthermore, the nearest neighbour approach involves assigning the digital number, which corresponds to the value of the closest original pixel, to the new pixel. This methodology retains the whole of the spectrum information, making it suitable for the efficient categorization of images (Harika et al., 2012).

3.2 Classification and land cover mapping

The land cover classification process employed the supervised maximum likelihood classification algorithm (MLC), which is widely recognized in the literature (Srivastava et al., 2012; Churches et al., 2014; Rawat and Kumar, 2015; Mubako et al., 2018). Additionally, a post-classification change detection analysis approach was utilized. The classification technique included a series of sequential phases. These steps included the identification of relevant characteristics and the selection of appropriate training regions. Subsequently, the training signature statistics and spectral patterns were thoroughly evaluated and analysed. Finally, the images were classified based on the procedures. In order to create training and validation samples, the use of Google Earth images was applied. The training and validation samples for the years 2017 and 2022 were obtained from Google Earth images, which were used as reference data for the imaging (Figure 2). The selection of training samples for each land cover class was based on considerations of representation and distribution. The data were stratified based on land cover classes and then used to create sample polygons of different sizes (3×3, 5×5, or 7×7 pixels), which were determined by the level of homogeneity within each class. In

order to include all land cover categories found in the research region, a total of 420, 460, and 460 sample polygons were taken into account for the years 2017, 2019, and 2022 (Table 1), respectively. Out of the total sample polygons, a proportion of one-third was selected at random for the purpose of developing and classifying signatures, while the remaining polygons were used for the purpose of assessing accuracy. The primary objective of this research was to conduct a comprehensive mapping of seven prominent land cover categories within the designated study region. The documenting of land cover features in the study conducted by El-Kawy et al. (2011) was not exclusively reliant on the used approach. It also integrated the first author's pre-existing knowledge and other data from sources such as topographic maps, visualizations from Google Earth, and interpretations.

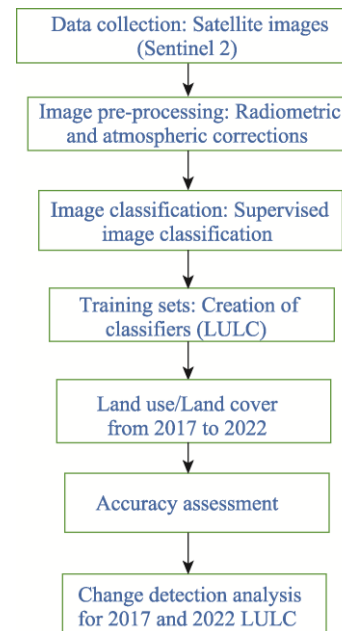


Figure 2 Methodology flowchart for LULC

Table 1 Source data for LULC in Annamayya District, Andhra Pradesh, India

Satellite	Sensor	Resolution	Source	Date of image	Link
Sentinel-2	MSI	10 m	Copernicus Open Access Hub	2017-09-08; 2018-09-15; 2019-09-10; 2020-09-14; 2021-09-12; 2022-09-26	https://scihub.copernicus.eu/

3.3 Change detection

For the assessment of changes in land cover spanning three decades, post-classification comparison techniques were employed. This choice was made due to its appropriateness and advantages in effectively illustrating the nature of change (Mas, 1999). By subtracting the classified maps from three specific time spans: 2017–2018, 2019–2020, and 2021–2022, alterations in land cover were charted (Figure 2). This methodology offers a depiction of changes from one state to another, which is recognized as one of the most

prevalent approaches for change detection (Skidmore et al., 1997; Sinha and Kumar, 2013; Alqurashi and Kumar, 2014; Ghosh et al., 2016). Examples of these changes include shifts from vegetation to agricultural land, transformation from open land to agriculture, and the conversion of water bodies to agricultural use, among other transitions. Alongside the commonly employed change detection methods, we also utilized time series analysis to measure changes in classified categories. This allowed us to monitor urban expansion, the expansion of agricultural land, and alterations in catchment vegetation throughout the study duration.

4 Results and discussion

4.1 Temporal monitoring of land cover

The images included in this study were acquired throughout the period of increased precipitation, namely from August to January, in each calendar year. The study analysed changes in land cover by using area data obtained from classified maps that provided information on various land cover categories, as shown in Table 2. The study region exhibited seven discrete land cover classifications, and their spatial arrangement for the period from 2017 to 2022 is shown in Figure 4. During the period of study from 2017 to 2022, a reduction of roughly 12.58% was seen in the extent of barren/degraded land. The area of barren or deteriorated land had a decline from 4912.45 km² in 2017 to 3848.72 km² in 2022, indicating a reduction of 1063.73 km². In the year 2017, the distribution of degraded lands was found to be evenly dispersed across the designated research area. In 2017, the district's complete geographical extent was ini-

tially occupied by a luxuriant forest cover, which accounted for around 659.48 km² (7.80%) of the area. By the year 2022, the aforementioned measurement had seen a growth to 1292.38 km², representing a percentage rise of 15.28% over a period of six years. The time span from 2017 to 2022 saw a significant rise in the extent of agricultural land, reaching around 228.93 km², equivalent to 2.71% of the total area. It is evident that human activities are the primary driver behind these changes, as there has been a notable growth in cultivation practices within the area. From 2017 to 2022, there was an observed increase of 33.05 km² around open water bodies (Figure 3). The observed disparity in forested regions may be clearly ascribed to the rise in conservation efforts and the substantial precipitation levels seen in 2022 (Figure 3). Winter crops were planted over the period spanning from November to February, which aligns with the timeframe in which the photographs were acquired. The precise temporal synchronization enabled the precise categorization of agricultural zones.

Table 2 Land use/Land cover from 2017 to 2022 of the study area

Land use/Land cover Classes	Area (km ²)					
	2017	2018	2019	2020	2021	2022
Waterbodies	42.74	63.89	64.22	68.14	72.11	75.79
Forest	659.48	736.15	381.49	674.85	1039.92	1292.38
Flooded vegetation	0.22	0.28	0.03	0.72	10.44	8.92
Agriculture land	2611.79	2735.71	2316.01	2662.8	2979.86	2840.72
Built-up land	191.68	227.97	252.26	273.86	302.87	348.66
Vegetation/Shrub	40.64	35.22	31.46	33.76	38.64	43.81
Barren/Degraded land	4912.45	4659.78	5413.53	4744.87	4087.27	3848.72

The present study represents the inaugural research of the spatio-temporal dynamics of LULC changes within the Annamayya district of Andhra Pradesh, India, spanning the period from 2017 to 2022. The present investigation used a combination of remote sensing data interpretation and field validation techniques to provide geospatially explicit data. The investigation substantiated noteworthy patterns, including the decline in forest resources and the proliferation of agricultural activities within the designated research region. Deforestation has been identified as the most severe kind of land cover change within the spectrum of degradation. Although tropical forests only occupy less than 7% of the Earth's surface, they are home to more than half of all plant and animal species (Mayers, 1992). Nevertheless, the phenomenon of tropical deforestation is resulting in significant species loss and exerting a detrimental influence on biodiversity by means of habitat degradation, the fragmentation of continuous forest areas, and the negative consequences associated with the edges and buffer zones that separate forests from deforested regions (Prance, 1982; Pimm et al., 1995).

Shifting agriculture and uncontrolled logging are identified as the main factors contributing to deforestation in this particular location. The process of deforestation does not occur in a linear fashion; rather, it exhibits complex routes and changes throughout several phases. The practice of shifting agriculture, also referred to as Podu cultivation, entails the clearing of small sections of deep forest by tribal tribes for the purpose of cultivation. These areas are used for a period of around 3 to 4 years, after which the inhabitants relocate to fresh territories for further cultivation. The activity has led to the transformation of formerly dense forest areas into unproductive terrain, especially as a consequence of the growing occurrence of shifting agricultural practices within these specific forested regions. The present integrated method provides a thorough examination of the LULC process, offering insights into the ongoing forest dynamics within the studied area. The establishment of a foundational database facilitates the conduct of more comprehensive and in-depth investigations in the future. Deforestation in tropical nations, such as India, is often associated with socio-economic issues, notably agricultural development

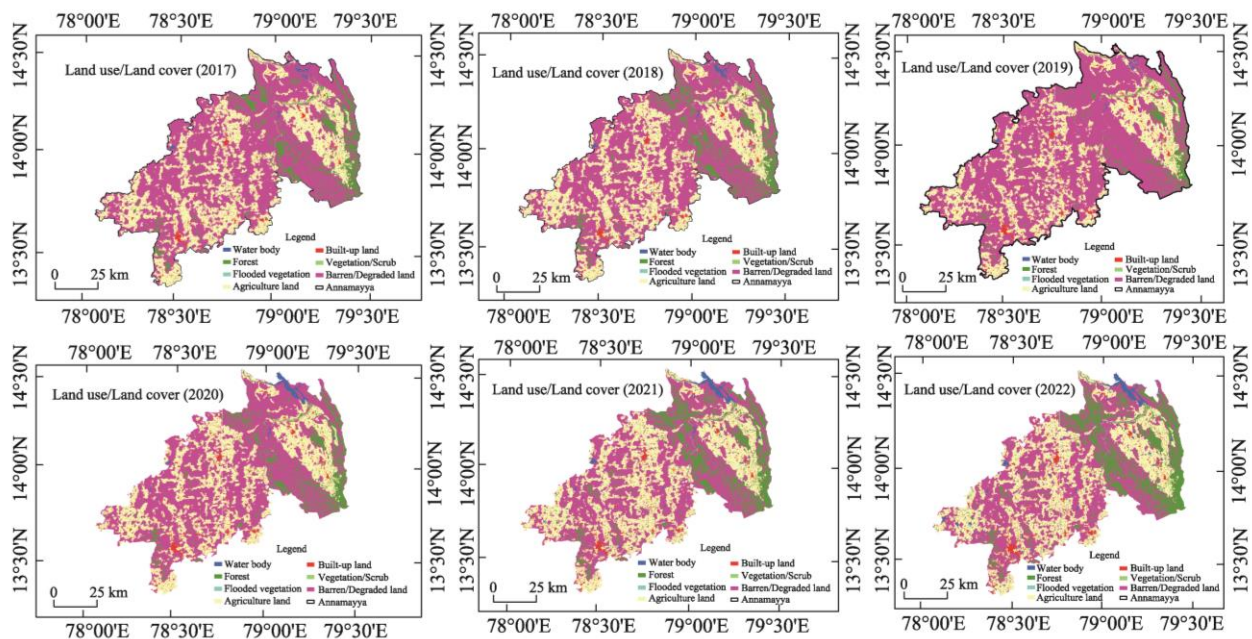


Figure 3 Land use/Land cover from 2017 and 2022 of Annamayya District, Andhra Pradesh, India

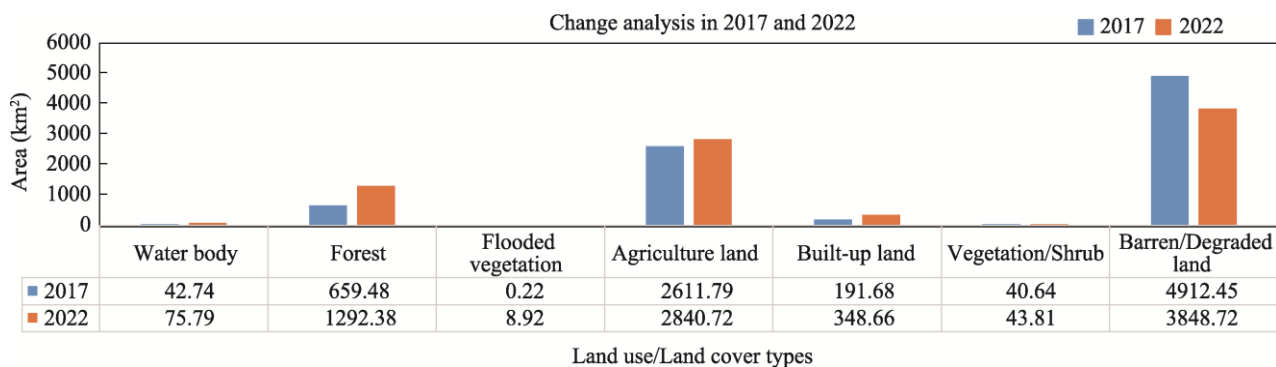


Figure 4 The Land use/Land cover Transition from 2017 to 2022

and population increase. Additionally, the establishment of road networks and communication infrastructure has been linked to elevated rates of deforestation. A multitude of landscape ecological investigations have shown the impact of socio-economic variables, forestry growth, urbanization, and agricultural patterns on forest landscapes (Pattanaik et al., 2011; Langat et al., 2021). Nevertheless, the factors contributing to land cover change are multifaceted and include a confluence of variables. In their study, Geist and Lambin (2001) undertook a meta-analysis of instances of tropical deforestation, revealing that a multitude of factors contribute to the broader phenomenon of land cover alteration. In order to broaden the scope of this research, it is recommended that same methodologies be used in other regions of Orissa to evaluate the extent of deforestation on a larger scale (Jiang and Tian, 2010; Areendran et al., 2013; Li et al., 2021). The use of automated satellite image categorization algorithms has shown promising progress in mit-

igating interpretation bias. Furthermore, the Markovian model has potential for doing exploratory investigation and making predictions about future events.

The results of this research provide significant insights into the patterns of deforestation and the factors that contribute to it. Significantly, the observed variations in the LULC process over the evaluated time frame (2017–2022) emphasize the need of carefully choosing a suitable reference period for accurately measuring deforestation. Incorporating immediate temporal intervals into the analysis has the potential to enhance the precision of change dynamics information, hence facilitating the development of a range of scenarios. In order to effectively address the challenges in the study area, it is imperative to adopt a strategic strategy that encompasses long-term forest protection, comprehensive management plans, active community involvement, and meaningful stakeholder participation. The identification of proximal determinants and driving forces of change that

influence the LULC patterns is an essential subsequent undertaking for this research (Petit and Lambin, 2002; Jiang and Tian, 2010; Areendran et al., 2013; Choudhary and Pathak, 2016; Nazrul et al., 2019; Li et al., 2021; Zhang et al., 2021).

4.2 Change detection analysis (2017 and 2022)

Land use and land cover (LULC) change detection analysis is a crucial method for understanding the dynamic processes affecting our environment. By analyzing changes in land cover over time, researchers and policymakers can assess the impacts of human activities, natural disasters, and climate change on the landscape. The table provided lists various LULC classes and their respective areas in km². The classes include waterbodies, forest, flooded vegetation, agriculture land, built-up land, vegetation/shrub, and barren/degraded land. The analysis of these changes can offer significant insights into environmental health and sustainability.

The data indicates a substantial presence of forested areas, covering 632.9 km². This dominant class suggests that forest ecosystems play a vital role in the region's ecology, providing habitat, maintaining biodiversity, and offering various ecosystem services. However, the negative value for Barren/Degraded Land (−1063.73 km²) requires further investigation. This anomaly might indicate a data entry error, or it could represent a significant reclamation or reforestation effort that has transformed previously degraded lands into more productive uses. The areas of waterbodies (33.05 km²) and agriculture land (228.93 km²) are also notable. Waterbodies are essential for supporting aquatic ecosystems, agriculture, and human consumption. Any change in their extent can directly affect water availability and quality. The considerable area of agricultural land highlights the importance of farming to the local economy and food security. Monitoring changes in agricultural land can reveal trends in crop production, land management practices, and potential pressures on agricultural sustainability.

Built-up land, with an area of 156.98 km², signifies urbanization and infrastructural development. The increase in built-up areas often corresponds with population growth and economic development. However, it can also lead to habitat loss, increased surface runoff, and environmental pollution. Managing urban expansion while balancing environmental conservation is a critical challenge for sustainable development. Lastly, the presence of flooded vegetation (8.70 km²) and vegetation/shrub (3.17 km²) indicates areas that may be susceptible to seasonal flooding or are transitioning land covers. Flooded vegetation can serve as vital wetlands that support diverse species and offer flood mitigation benefits. However, changes in these areas need to be monitored to understand hydrological cycles and the impacts of climate variability. The LULC change detection analysis from the provided data reveals the complex interplay between natural

landscapes and human activities. The implications of these changes are far-reaching, affecting biodiversity, ecosystem services, climate regulation, and socio-economic development. By continuing to monitor and analyse LULC changes, stakeholders can make informed decisions to promote sustainable land management and mitigate adverse environmental impacts.

5 Conclusions

The study conducted from 2017 to 2022 revealed significant transformations in land resource utilization in Annamayya district, Andhra Pradesh, India. Key findings include a substantial reduction of approximately 12.58% in barren or degraded land, from 4912.45 km² in 2017 to 3848.72 km² in 2022. During the same period, the forest cover increased markedly from 659.48 km² (7.80%) to 1292.38 km² (15.28%), indicating a significant improvement in forest conservation and precipitation levels. Agricultural land also expanded by 228.93 km² (2.71%), driven by intensified cultivation activities. Additionally, the area around open waterbodies grew by 33.05 km². These changes underscore the crucial role of human activities and conservation efforts in shaping land use patterns. The study highlights the importance of utilizing satellite remote sensing and GIS to monitor these changes, providing valuable insights for resource managers to optimize land resource management and develop strategies that account for ecological impacts on biodiversity.

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基于遥感和 GIS 分析的土地利用/覆盖变化的案例研究：以印度 Annamayya 区为例

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摘 要: 要了解和管理不断变化的景观, 需要分析土地利用和土地覆盖 (LULC) 的变化。本研究结合遥感数据和 GIS 技术, 揭示这些变化背后错综复杂的模式和驱动因素。通过整合 GIS 分析, 我们确定了导致这些变化的因素, 例如城市化、农业扩张、森林砍伐和自然资源管理。该研究基于 2017–2022 年的土地利用/土地覆被变化数据 (特别是 Sentinel-2 图像数据), 利用 GIS 工具和卫星数据解译创建了数字化 LULC 图。土地利用与土地覆被类型分为 6 种: 农业用地、建设用地、荒地或退化土地、被淹没的植被区、森林、植被或灌木覆盖区和水体。研究发现, 在 2017 年至 2022 年间, 研究区荒地面积减少了 1063.3 km²。农业用地扩展了 228.93 km², 林地增加了 632.90 km², 水体面积增长了 33.05 km²。这些变化表明, 2017 年至 2022 年间高强度降雨可能影响了区域土地利用与土地覆被的转换。研究结果对土地管理、环境保护和区域的可持续发展具有重要意义。

关键词: 土地利用/土地覆盖; 遥感; GIS; 转变; 环境; 可持续发展